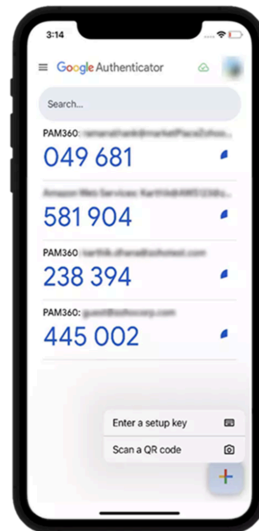


Mobile Application Design Blueprint: Google Authenticator

The design framework of Google Authenticator follows a utility-first mobile paradigm tailored for high-precision, rapid micro-interactions. Unlike engagement-driven apps, multi-factor authentication (MFA) client apps prioritize minimal cognitive load, immediate clarity, and zero-error code extraction.



UX Philosophy & Core Principles

Authentication interrupts primary user goals; thus, the UI minimizes extraneous visual noise.

- **Temporal Visibility:** 30-second Time-based One-Time Passcode (TOTP) cycles use an ambient countdown timer, preventing expired code submissions.
- **Cognitive Ergonomics:** Passcodes use monospaced fonts and XXX YYY 3+3 digit chunking to reduce mental friction and typing errors.
- **Instant Feedback:** Single-tap copy operations trigger haptic feedback and explicit contextual toast notifications confirming the active account.

Visual Screen Architecture

1. **Main Account Vault:** A vertical list displaying account cards. Each card contains the service issuer name, user identifier, a 6-digit TOTP code, and a circular radial progress ring depleting over 30 seconds. A prominent Floating Action Button (FAB) anchored at the bottom-right opens onboarding options.
2. **Setup Viewfinder & Form:** A full-screen camera overlay with a square targeting box for scanning otpauth:// QR codes, backed by a manual Base32 key entry form fallback.

Process Flows & Human Factors Engineering

- **Onboarding Flow:** Tap FAB -> Select QR scan or manual key -> Optical framing / Base32 input -> Parse secret & render token card.
- **Retrieval Flow:** Launch app -> Instant list view -> Single tap card to copy token -> Clipboard update with haptic/toast confirmation.